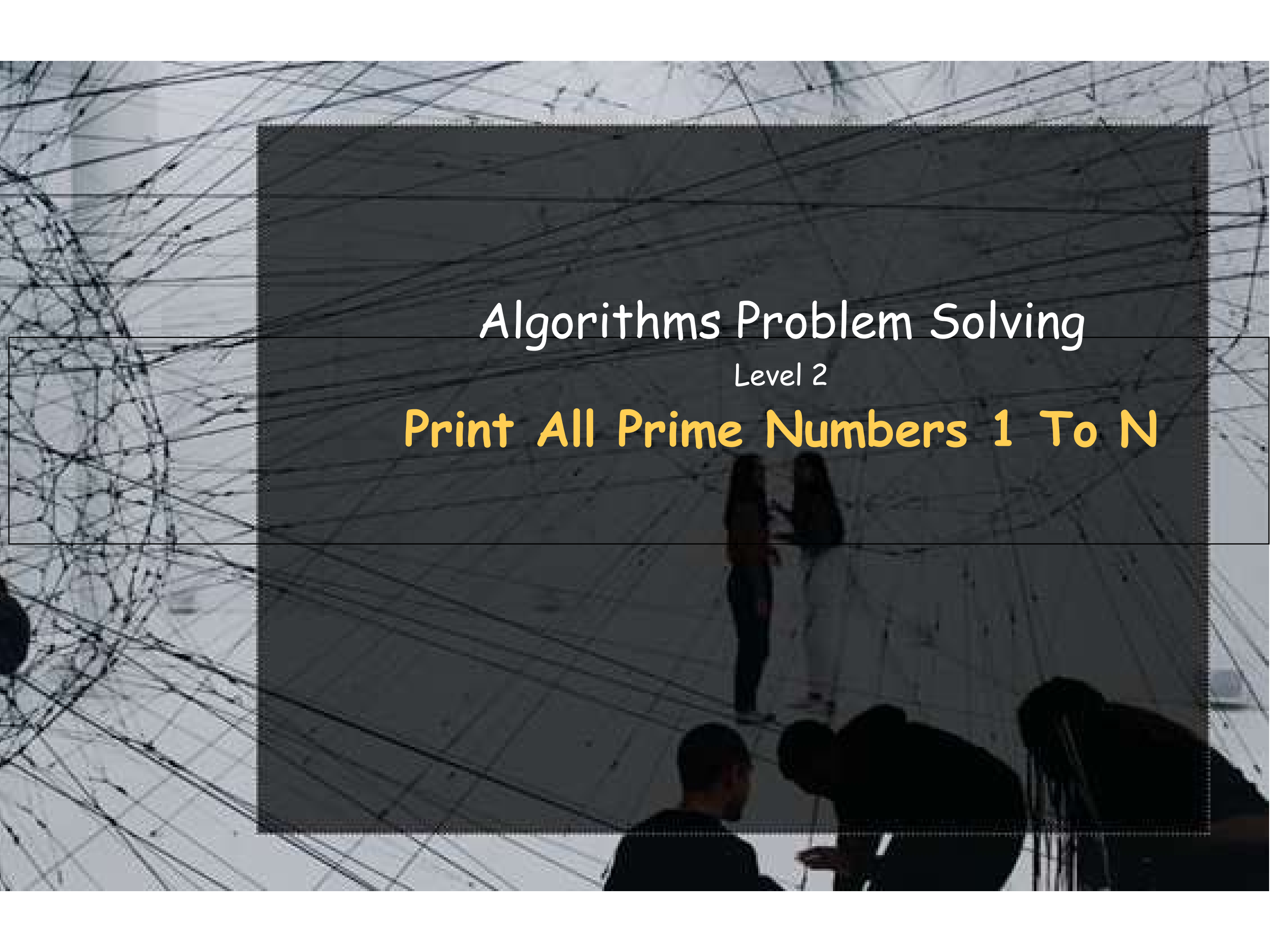


Algorithms Problem Solving

Level 2

Print All Prime Numbers 1
To N

The background of the slide is a photograph of a modern building's interior. It features a large, curved glass ceiling that allows natural light to filter through, creating a pattern of light and shadow on the floor. In the lower portion of the image, the silhouettes of several people are visible, some standing and others sitting, suggesting a collaborative workspace or a public area. The overall color palette is muted, with greys, blues, and the warm tones of the light coming through the glass.

Algorithms Problem Solving

Level 2

Print All Prime Numbers 1 To N

Definition

- ◆ A prime number is a number greater than 1 that can only be divided exactly by 1 and itself.
- ◆ This program reads a positive number N and prints all prime numbers from 1 to N .

Explanation

- ◆ ``ReadNumber()`` reads a positive number from the user using a ``do-while`` loop.
- ◆ ``CheckPrime(int numb)`` checks whether a number is prime or not.
- ◆ If the number is less than 2, it is ``NotPrime``.
- ◆ The ``for`` loop checks possible divisors starting from 2 up to ``numb / 2``.
- ◆ The ``%`` operator checks whether the number is divisible by another number.
- ◆ If ``numb % i == 0``, the number has another divisor, so it is ``NotPrime``.
- ◆ If no divisor is found, the function returns ``Prime``.
- ◆ ``PrintPrimeFrom1ToN()`` loops from 1 to N and calls ``CheckPrime()`` for every number.
- ◆ Only numbers whose result is ``Prime`` are printed.

Steps in main()

- 1 Read a positive number N.
- 2 Start a loop from 1 to N.
- 3 Check each number using `CheckPrime()`.
- 4 If the result is `Prime`, print the number.
- 5 Continue until reaching N.

💡 Example

◆ Input:

`10`

◆ The program checks:

`1 → NotPrime`

`2 → Prime`

`3 → Prime`

`4 → NotPrime`

`5 → Prime`

`6 → NotPrime`

`7 → Prime`

`8 → NotPrime`

`9 → NotPrime`

`10 → NotPrime`

➡ Result

Prime numbers From 1 to 10 are :

2

3

5

7

Note

- ◆ `1` is not a prime number because a prime number must be greater than 1.
- ◆ The enum `enPrimeOrNot` makes the result more descriptive than returning a simple `true` or `false`.

⊘ Warning

- ◆ The comment in the original code says the output for `10` includes `1`, but that is incorrect. The actual program correctly excludes `1`.
- ◆ `CheckPrime()` checks divisors only up to `numb / 2`. This works correctly, although checking up to the square root of the number would be more efficient for larger numbers.



Comparison

Prime

- ✓ Greater than 1.
- ✓ Divisible only by 1 and itself.
- ✓ Example: `2, 3, 5, 7`.

Not Prime

- ✓ Less than 2, or
- ✓ Has another divisor besides 1 and itself.
- ✓ Example: `1, 4, 6, 8, 9, 10`.

✦✦ Summary

- ✓ Read a positive number N.
- ✓ Check every number from 1 to N.
- ✓ Use `%` to test divisibility.
- ✓ Numbers with no divisor other than 1 and themselves are prime.
- ✓ `1` is not prime.
- ✓ The enum `Prime / NotPrime` describes the checking result clearly.

Thank you 😊